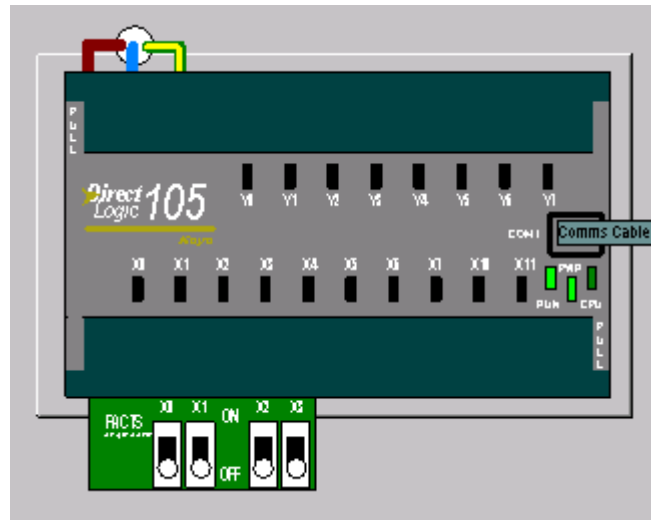


Koyo Direct Logic KDL105KS Protocol driver

Overview

Adroit supports communications with the Koyo Direct Logic DL105 PLC Via the K-Sequence protocol.



Device configuration

KDL105KS Protocol Driver : BOILER1

Koyo Direct Logic 105 Details

CPU Model: DL105

Station ID: 1

Port details

Comm Port: COM1

Baud rate: 9600

Parity: Odd

Stop Bits: One

Retries: 3

Timeout (ms): 1000

OK

Cancel

Help

DL105 Details

PLC Station ID: Fixed at 1

Port Details

Port: The available serial ports on your machine

Baud Rate: The speed of data transmission. Fixed at 9600

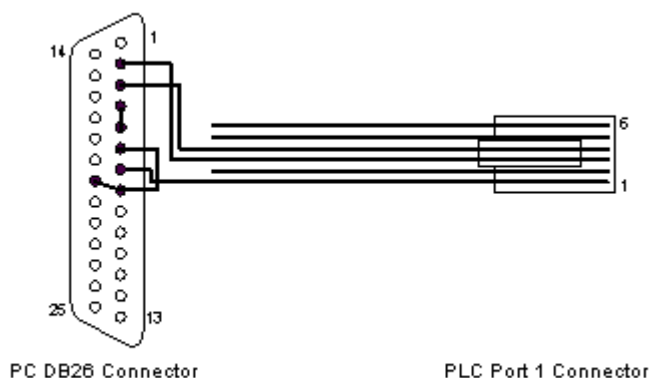
Parity: Check bit. Fixed at Odd.

Stop Bits: UART signal bits. Fixed at 1.

Retries: The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Timeout: The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Wiring



The diagram above shows the wiring for the comms cable between a PC serial port and the DL105 Com 1 port.

Supported addresses

Prefix	Start	End	Global Start	Global End	Type	Access	Description
TA	0	77	V0	77	WORD	Read Write	Timer Accumulator
CTA	0	77	V1000	1077	WORD	Read Write	Counter Accumulator
			V2000	2377	WORD	Read Write	Variable
			V4000	4177	WORD	Read Write	Variable
			V7260	7647	WORD	Read Write	System Status
			V7750	7777	WORD	Read Write	System Status
X	0	177	V40400	40407	BIT	Read Only	Input
Y	0	177	V40500	40507	BIT	Read Write	Output
C	0	377	V40600	40617	BIT	Read Write	Variable
S	0	377	V41000	41017	BIT	Read Write	Stage Status
T	0	77	V41100	41103	BIT	Read Write	Timer Status
CT	0	77	V41140	41143	BIT	Read Write	Counter Status
SP	0	117	V41200	41204	BIT	Read Only	System Status
SP	540	577	V41226	41227	BIT	Read Only	System Status

Adroit Data Types

Supported Data types are as follows:

PLC Data Type	Adroit Data Type		
	Bool	Integer	Real
WORD Read		√	√
WORD Write		√	√
BIT Read	√	√	√
BIT Write	√	√	√

Driver Implementation

NB All addresses with the Koyo DL105 PLC are expressed in Octal.

When reading or writing data to or from the DL105 the driver is actually always using 16 bit words. The driver, depending on the prefix, will work out the type of data and get or put only the correct information. For example when a digital input “X” prefix is used it can be mapped to any Adroit data type. The global equivalent for X0 is V40400. This is a word and X0 to X20 (The next 16 bits) are stored here. This can be read into any Adroit data type except BOOLEAN or any of its derivatives. The most efficient way of reading or writing digitals is to use a multistate agent and read the global equivalent address.

Troubleshooting

During the installation of the driver the following files will be copied:

KDL105KS.DLL	The driver itself.
KDL105KS.PDF	The driver help file.
KDL105KS.WGP	An Adroit database file.
KDL105KS.MIM	An adroit mimic file.
KDL105KS.REG	Registry file

Once the driver has been installed and setup, run the REG file, you may then set the Adroit database file to the KDL205KS.WGP and start the agent server.

When the agent server is ready, start the user interface and open the file KDL105KS.MIM. You will then see a representation of a DL105 base with an input simulation module.

If all is installed correctly and the driver has successfully started, you will be able to see the inputs change when the simulator module switches are changed and also will be able to change output states when the output lights are highlighted and clicked.

This is the first thing to run when first setting up the unit. It can also be used as a training aid to see how the unit works.

Revision Information

Date	Changes
11-Jun-02	Adroit Technologies takes over driver from Alex Griffiths.
14-Aug-03	PDF format document distribution from now on